EVERY MODULE IS A DIRECTED COLIMIT OF FINITELY PRESENTED MODULES

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Allowed standard prerequisites (editorial).

- Finite presentations as cokernels of maps between finite free modules. Used to define the modules M_{S,E} in the counted proof (lines 1856-1864). Produces finitely presented modules from a finite set of elements and a finite set of relations.
- Directed colimits indexed by a poset of finite data. Used at the end of the counted proof (lines 1866-1872). Organises the finitely presented approximations into a directed system whose colimit is M.

Lemma 0.1. *Let R be a ring and let M be an R-module. Then M is the colimit of a directed system (M_i, μ_{ij}) of R-modules with all M_i finitely presented R-modules.*

Proof. Consider any finite subset $S \subset M$ and any finite collection of relations E among the elements of S. So each $s \in S$ corresponds to $x_s \in M$ and each $e \in E$ consists of a vector of elements $f_{e,s} \in R$ such that $\sum f_{e,s} x_s = 0$. Let $M_{S,E}$ be the cokernel of the map

$$R^{\#E} \longrightarrow R^{\#S}, \quad (g_e)_{e \in E} \longmapsto (\sum g_e f_{e,s})_{s \in S}.$$

There are canonical maps $M_{S,E} \to M$. If $S \subset S'$ and if the elements of E correspond, via this map, to relations in E', then there is an obvious map $M_{S,E} \to M_{S',E'}$

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commuting with the maps to M. Let I be the set of pairs (S, E) with ordering by inclusion as above. It is clear that the colimit of this directed system is M. $\square$